# 4

# Fat

## INTRODUCTION

From a nutritional perspective, "fat" is a generic term to describe dietary compounds that are predominantly fatty acids (FAs), which include triglycerides (TGs), phospholipids, galactolipids, nonesterified FAs, and salts of FAs. The glycerol backbone in TGs and some phospholipids is nutritionally equivalent to a carbohydrate, as are the sugar moieties in galactolipids. These FA compounds are soluble in nonpolar solvents, and traditionally, crude fat (CF) or ether extract has been defined based on gravimetric determination of lipid-extractable com- ponents of feeds. Because the solvent used is not always truly an ether, and various modifications of the chemical procedure exist, the term "crude fat" is used in this publication, rather than "ether extract," to define lipid mass in feeds determined gravimetrically. This solvent-extracted mass includes the non-FA portion of the crude fat, including moieties covalently bound to FA as well as non-FA lipids. Fat is typically fed to increase the energy density of the diet, and long-chain FAs are the major energy-rich moiety of fats. Therefore, specifying dietary FA content of feeds is preferable to extraction method- ology, especially because different extraction methodologies, such as ethyl-ether extraction, hexane or "petroleum ether" extraction, or acid hydrolysis ether extraction, yield different values. Additionally, determination of total FA in feeds using quantitative recovery and subsequent analysis by gas-liquid chromatography gives useful information on the specific FA fed. Quantitative extraction of lipids prior to FA analysis is critical, and internal standards of FA that do not occur in feeds should be added to adjust for loss of FA during the analytical process. However, when extraction of lipids is followed by subsequent FA analysis, there is no concern about using solvent systems that also extract non-FA components, provided these compounds do not subsequently manifest as unidentified FA in the chromatogram and mistakenly contribute to total FA. See Chapter 18 for additional details on methodology.

The primary dietary FAs are the 16-carbon and 18-carbon saturated FAs, palmitic acid (Cl6:0) and stearic acid (Cl8:0), and the unsaturated 18-carbon FAs with cis double bonds: either a single double bond at the 9 carbon (oleic acid, C18:1 cis-9), double bonds at the 9 and 12 positions (linoleic acid, C18:2 cis-9, 12), or double bonds at the 9, 12, and 15 positions (the "alpha" form of linolenic acid, Cl 8:3 cis-9, 12, 15). Other "cis" unsaturated FAs, such as palmitoleic (C16:1 cis-9) and gamma-linolenic acid (C18:3 cis-6, 9, 12), gener- ally are not major components of mixed diets but may occur at significant levels in certain feeds. Elaidic acid (Cl8:1 trans-9) and other trans-C18:l FAs may also be present in diets. Vaccenic acid (C18:1 trans-11) is present naturally in unprocessed tallow, and elaidic acid and other trans-C18:1 FAs are prevalent in partially hydrogenated fat supplements and may be present in other feeds. Also, FAs of 20 carbons or more can be found in feeds such as rapeseed and marine products. The list is not so vast as to preclude individual FA analysis, especially because all are easily quantified by the same gas-liquid chromatography method. As more feeds are analyzed by chromatography, near-infrared spectroscopy prediction equations might be able to provide information on FA content. If FA concentration is not known, it can be estimated from crude fat data (Daley et al., 2020; see Chap- ter 19). Within, but not across, a specified feed source, the fraction of crude fat that is FA, and the proportion of specific FA within total FAs, often is reasonably constant. However, many domestic plant species have distinct strains that vary markedly in FA proportions as a result of selective breeding or by direct genetic modification.

The term "fat" can refer to TGs in a solid form, but "oil" usually refers to TGs containing more unsaturated FAs and are liquid at 25°C. Oleic, palmitic, and stearic acids have melting points of 7°C to 16°C, 63°C, and 71°C. Salts of FAs have higher melting points that are dependent on the cation. Adding liquid oils or molten fats to feeds has the ad- ditional value of reducing dustiness of feeds. High melting point fats manufactured into dry, granular products can be handled easily in conventional feed systems, and this can be a significant advantage provided they remain nutritionally available. When typical forages and grain concentrates are fed to cattle, dietary FA concentration is near 3 percent of diet dry matter (DM), but all-forage diets and early growth pasture can have significantly higher levels of FAs (Boufaied et al., 2003). Various forms of supplemental fat can be fed, including oilseeds, rendered animal fats, extracted plant oils, and processed dry, free-flowing fats. In addition to increas- ing energy density of the diet, fat supplementation can also increase absorption of fat-soluble nutrients and provide es- sential FAs (Jenkins and Harvatine, 2014). In general, diets with greater than 7 percent total dietary FAs are not recom- mended (NRC, 2001).

## RUMEN METABOLISM, DIGESTION, AND ABSORPTION

For general reviews of lipid digestion and absorption in ruminants, see Noble (1981) and Jenkins (1993), as well as the quantitative reviews of FA digestion of Glasser et al. (2008b), Schmidely etal. (2008), and Boerman et al. (2015a). Dietary esterified FAs are rapidly hydrolyzed by lipolytic mi- croorganisms within the rumen to yield free FAs. Following hydrolysis, individual unsaturated FAs can be hydrogenated by ruminal bacteria. Complete hydrogenation of an unsatu- rated FA involves several steps that may be performed by dif- ferent microbial species (Dewanckele et al., 2020); therefore, the extent of biohydrogenation and the products of hydroge- nation vary. Stearic acid formed by complete biohydrogena- tion of unsaturated dietary Cl8 FA is produced by ruminal microbes in intimate contact with an unsaturated FA molecule that is not buried inside other FA molecules. These exposed FAs are dispersed on feed particles or are part of the micro- bial flow and are not thought to physically reaggregate. This distributed stearic acid may have a different digestive fate than stearic acid or stearate salts aggregated into larger particles in concentrated fat supplements. Biohydrogenation of linoleic or linolenic acid reduces their postabsorptive ability to meet the animal's requirement for these two essential FAs.

Incomplete biohydrogenation of linoleic and linole- nic acid can result in a variety of FAs, including various conjugated linoleic acids (CLAs) and trans-monoenoic FAs (Bauman and Griinari, 2003; Bauman et al., 2011). Trans-monoenoic FA can also arise from isomerization of oleic acid. Some bacteria hydroxylate FAs, but these routes probably represent a minority of FA transformations in the rumen (Fulco, 1974; McKain et al., 2010). Microbial produc- tion of trans-10 FA is increased in diets with higher starch fermentability and ruminal lactate concentration. Several of these microbially produced FAs have bioactivity and health implications in both the animal and humans eating the fat derived from these animals (Bauman and Lock, 2006). Most dramatically, trans-10, cis-12 CLA directly reduces mammary gland fat secretion in lactating cows. Other CLAs have also been shown to be active, but cis-9, trans-11 CLA, which is formed largely from mammary gland desaturation of absorbed trans-11 Cl 8:1, does not depress milk fat secre- tion (Bauman and Lock, 2006). Elevated milk trans-10 C18:1 has a strong statistical association with milk fat depression (Matamoros et al., 2020) and may (Shingfield et al., 2009) or may not (Lock et al., 2007) have a direct causal effect on mammary fat secretion. If trans-10 FA is bioactive, it is much less potent and required at higher concentrations than the active CLA isomers. However, trans-10 FA also frequently occurs in much greater quantities in milk fat than those CLAs that depress milk fat secretion.

Estimates for net ruminal disappearance of polyunsatu- rated FAs (PUFAs), presumably via biohydrogenation, range from 60 to 90 percent (Bickerstaffe et al., 1972; Mattos and Palmquist, 1977; Jenkins and Bridges, 2007). Because of hydrogenation in the rumen, Cl8:0 and Cl8:1 are the major FAs leaving the rumen. Some of the 18:1 leaving the rumen is cis-9 (oleic) but also includes trans-10, trans-11, and other monoenoic 18-carbon FAs derived from isomeriza- tion of dietary oleic or partial hydrogenation of linoleic and linolenic acids (Glasser et al., 2008b). Loss of oleic, linoleic, and linolenic acids in "unprotected" feed fats was 86, 82, and 86 percent, respectively (Jenkins and Bridges, 2007). This extensive ruminal loss of PUFA was true for oilseeds and for calcium (Ca) salts, but duodenal oleic flow was often increased with oilseeds or "protection." Data showing formaldehyde treatment effects to increase duodenal passage of unsaturated FA to the duodenum are very limited, but formaldehyde treatment does appear to increase transfer of dietary linoleic and linolenic, as well as oleic, into milk and body fat (Jenkins and Bridges, 2007). Because monoenoic FAs are an important amphiphile, this increased duodenal flow of oleic acid may be beneficial for FA digestion.

The ruminal escape of linoleic acid averages 20 percent with a range of 5 to 30 percent; for linolenic acid, corre- sponding values are 8 percent and 0 to 15 percent (Doreau and Ferlay, 1994). Noble (1984) reported estimates of pyloric flow of linoleic acid as 0.3 to 0.5 percent of diet intake on an energetic basis. Glasser et al. (2008b) reported a mean flow to the duodenum of total C18:2 of 2.3 g/kg dry matter intake (DMI), with a range of 0 to 12.7. In addition, they reported that linoleic acid ranged from 5.4 to 98.4 percent of total 18:2, with means of 65.3 percent and 80.3 percent in basal and lipid-supplemented diets, respectively. In this same review, 18:3 flow ranged from 0 to 3.5 g/kg DMI, with an average flow of 0.5. The linoleic acid requirement for growing and reproductive swine is set at a 1 g/kg intake (NRC, 2012). A dairy diet could easily contain 1 percent linoleic acid, so 10 percent apparent escape would just meet this requirement. Ca salts of rapeseed oil FAs did not protect these PUFAs compared to free rapeseed oil TG (Ferlay et al., 1993). Similar results were reported for soybean oil and Ca salts of soybean oil (Lundy et al., 2004). Small changes in mass flow of these essential FAs may have important effects on animal health, but these responses may be difficult to evaluate and the changes in FA flow difficult to produce or predict with current technology. The postabsorptive fate of PUFA following absorption is discussed in Lanier and Corl (2015).

In addition to their role as sources of essential FAs and duodenal conjugated linoleic and trans-monoenoic FAs, plant oils rich in unsaturated FAs have previously been purported to have a marked negative effect on fiber digestibility (Jenkins, 1993). Based on analysis of literature of fat effects on total- tract neutral detergent fiber (NDF) digestibility in lactating dairy cattle, Weld and Armentano (2017) found a marked negative effect of supplemental C12 and C14 saturated FAs (lauric and myristic) on total-tract NDF digestibility, a modest negative effect of vegetable oils, and no significant depressing effect of other FA sources. At 3 percent supple- mental FA from vegetable oils, the long-chain FA oil effect would correspond to a decrease of about 1.3 percentage units of digestible NDF. The response appears linear, so feeding a very high level of oils should be avoided. In the few studies that reported rumen NDF digestion, oil supplements did not appear to decrease rumen digestion. At typical fat supple- mentation levels, the negative effect of oils on digestible energy (DE) intake through depression of NDF digestion is minor compared to their positive effects on diet energy density and their potentially negative effects on DMI (Weld and Armentano, 2017). Although there was no significant effect associated with C:16 in Weld and Armentano (2017), subsequent studies have shown increased NDF digestibility when fats composed mostly of C16:0 are added to the diet (de Souza and Lock, 2019; Western et al., 2020).

Microbial oxidation of long-chain FAs is limited, although disappearance of FAs of 14 carbons and shorter occurs in the rumen in vivo (Wu et ah, 1991), although in vitro, loss is in FAs shorter than C14 (Wu and Palmquist, 1991). Net FA bal- ance across the rumen is also affected by FA synthesis by ru- men microorganisms, which is obvious at least for odd-chain and branched-chain FAs both in vitro (Wu and Palmquist, 1991) and in vivo (Vlaeminck et ah, 2006). Regression of FA flow at the duodenum versus intake revealed a slope of only 0.8 (g duodenal FA/g intake FA) and an intercept of 9.3 (g duodenal FA/kg DMI) with a common regression for “protected” and other lipids (Doreau and Ferlay, 1994). The intercept gives an estimate of net endogenous rumen synthesis of FA, and the slope of 0.8 indicates 20 percent true disap- pearance of FA in the rumen, assuming a constant endogenous FA synthesis as fat intake increases. Disappearance could be due to microbial catabolism, or catabolism or absorption by forestomach epithelium. A similar result was obtained when plotting FA duodenal flow (g/d) against intake of FA (g/d), with an intercept of 93 g/d and a slope of 0.84, corresponding to 16 percent true digestibility in the rumen (Boerman et ah, 2015a). If endogenous FA is set at 15 g/kg lipid free organic matter digested in the rumen and assumed to decrease linearly as FA intake increases, then the slope of estimated dietary FA passage relative to intake could be interpreted as a lower true ruminal digestibility of 8 percent (Jenkins, 1993). The quantitative relationship based on constant endogenous FA synthesis of 9.3 g/kg DMI and 20 percent true rumen disap- pearance of FAs would yield 0 percent FA apparent ruminal digestibility at 4.6 percent dietary FA, negative values (net ruminal synthesis of FAs with duodenal flow exceeding in- take) below that, and about 7 percent apparent digestibility in the rumen at 7 percent FA in diet DM.

Most FAs synthesized by rumen microbes are incorpo- rated into phospholipids. Approximately 85 to 90 percent of the FA leaving the rumen are free FAs, and 10 to 15 percent are microbial phospholipids. Because FAs are hydrophobic, they associate with particulate matter and pass to the lower gut with those particles.

Bile and pancreatic lipase are required for duodenal TG digestion and absorption. If TGs are fed at moderate levels in a form that protects them from rumen microbial hydrolysis (e.g., formaldehyde-protected casein-fat emulsion), suf- ficient lipase activity is present for triglyceride hydrolysis (Noble, 1981). However, pancreatic lipase does not appear to be inducible (Johnson et al., 1974) and may become limiting if large quantities of TG are presented to the small intestine. In the absence of substantial amounts of monoglyceride reaching or being formed in the small intestine, the mecha- nism for FA emulsification in ruminant is unclear but may involve lysolecithin and monounsaturated FAs (MUFAs) in addition to bile acids. Oleic acid and mono-olein are consid- ered important intestinal amphiphiles with a critical micellar concentration of 0.55 and 0.60 mM and saturation ratios of 1.04 and 1.70, but trans-9 18:1 and trails-11 18:1 FAs have critical micellar concentrations of 1.20 and 0.70 mM, as well as saturation ratios of 0.51 and 1.39, which makes them less nonpolar than palmitic and stearic acids, which have a critical micellar concentration of 1.80 and 1.40 mM and saturation ratios of 0.16 and 0.07 (Freeman, 1969). The frans-monoenoic FAs are present in duodenal chyme as a result of ruminal microbial action on dietary unsaturated FAs (Glasser et al., 2008b). Comparable values for linoleic acid are 0.35 mM and 1.04. Lysolecithin is formed by pancreatic phospholipase activity on lecithin from microbial or hepatic origin. MUFAs are predominantly from digesta leaving the rumen; therefore, increasing the flow of unsaturated FAs to the duodenum may improve FA digestibility. Intestinal infusion of an emulsi- fier can improve digestion of FA (de Souza et al., 2020). FA emulsification and micelle formation in the small intestine are essential for the efficient absorption of fat.

## MODEL USED FOR FATTY ACID DIGESTION

The mathematical model of total-tract FA digestibility applied in this revision is the simplest possible (other than fixed FA digestibility across all feeds) and likely oversimplifies the biology inherent in the digestion process while allowing differ- ent FA sources to be assigned different inherent digestibility. Total-tract digestibility, as opposed to intestinal, is estimated to be consistent with the energy model and to allow access to

TABLE 4-1 Calculated Total-Tract Digestibility Coefficients of FA

| Fatty Acid Source Class | Total-Tract Digestibility Coefficients[a] | Standard Error | Fatty Acid Composition (% of FA)[b] |
|---|---|---|---|
| Common feeds | 0.73 | 0.026 | — |
| Oil seeds | 0.73 | 0.041 | — |
| Oil | 0.70 | 0.033 | PUFA" >20%, UFA“' >65% |
| Blended triglyceride | 0.63 | 0.027 | PUFA <20%, UFA >56% |
| Tallow triglyceride | 0.68 | 0.029 | MUFA' >36%, UFA <56% |
| Saturated fatty acid enriched triglycerides | 0.61 | 0.037 | MUFA >25%, UFA <36% |
| Extensively saturated triglycerides | 0.44 | 0.030 | MUFA <20%, UFA <25% |
| Calcium salts palm fatty acid | 0.76 | 0.027 | MUFA >30% |
| Saturated fatty acid enriched nonesterified fatty acid | 0.69 | 0.022 | MUFA <15%, UFA <20% |
| Palmitic acid, -85% | 0.73 | 0.077 | — |
| Palmitic or stearic acid >90% | 0.31 | 0.046 | UFA <2% |

[a]Regression of apparently digested FA on FA intake yields a true digestion coefficient, but because the regression intercept for apparently digested FA is set to 0, true and apparent digestion are equal (Daley et al,, 2020).

[b]Classifications provide unique classes among the triglyceride supplements reported in this database by assigning the fat source to the first row it satisfies. MUFA = monounsaturated fatty acid; PUFA = polyunsaturated fatty acid; UFA = unsaturated fatty acid.

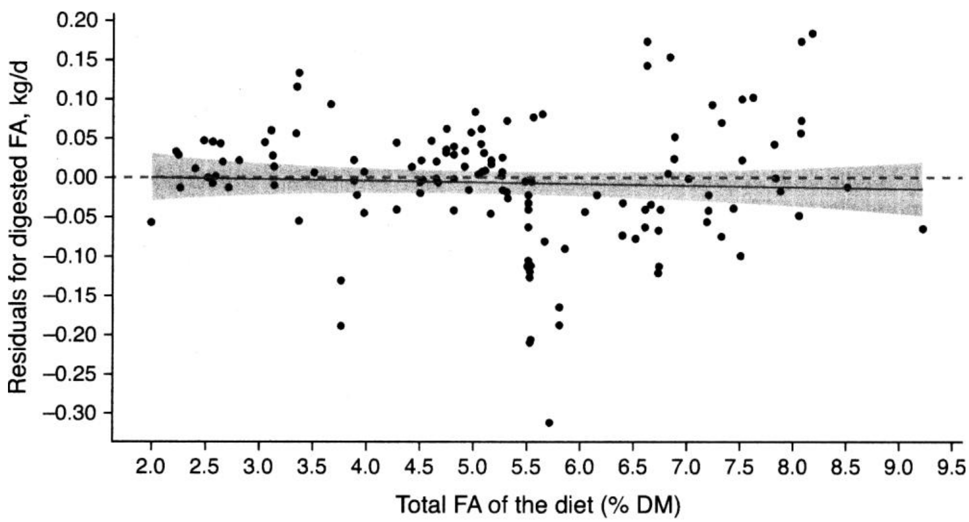


FIGURE 4-1 Residuals for final prediction model used for FA digestion. Concentration of dietary FA had a positive but nonsignificant (P = 0,65) estimate, indicating no depression in FA digestion with increased dietary FA concentrations. The intercept was 0.005, and slope was 0.002.
SOURCE: Figure and statistical information courtesy of V. L. Daley based on regression analysis from Daley et al. (2020).

the most data possible. The model applies constant digestibility of FA for a given feed independent of the amount of DM or FA fed. It assigns digestion coefficients to classes of feeds as estimated by multiple linear regression analysis (Daley et al., 2020). In this simple model, diet digestibilities are therefore assumed to be additive, with the digestibility of total diet FAs the weighted average of the individual feeds as is commonly done with static nutrients in ration balancing. This model used a meta-regression of apparently absorbed FA (across the entire tract) as a linear function of FA intake (i.e., a “Lucas” test) from distinct classes of FA sources as defined in Table 4-1. The slope parameter determined for each class is the estimated true digestibility for total FA for the feeds in that class. The intercept (FA absorbed at 0 FA intake) was small, statistically not different from 0, and positive (implying negative endogenous fecal FA, which is impossible). Therefore, the intercept (and endogenous FA secretion) was set to 0 in the final model. Inclusion of DMI, or total diet FA concentration as additional independent terms, was tested and found to be nonsignificant (V. L. Daley, National Animal Nutrition Program, personal communication, January 4, 2021). The residuals for the cho-sen equation showed no bias due to dietary FA concentration (see Figure 4-1). Because endogenous fecal FA has been set to zero, true and apparent digestibility are the same. Based on the regression equation, the default true digestibility of FA from most feeds was set at 0.73 (see Table 4-1). Digestibility of FA for individual feeds can be changed at the discretion of the user.

Because FAs are synthesized in the rumen and cell sloughing occurs along the digestive tract, some endogenous fecal FAs should be excreted. Therefore, setting endogenous FA secretion to zero is probably not biologically correct but was adopted based on regression fitting and parameter bounds within the realm of possibility. Fitting fecal FA (White et al., 2017, sup. 3) to a simple linear digestion model of FA resulted in a similarly impossible negative intercept for fe- cal FA at zero FA intake. Only by fitting an increasing slope function of fecal FA to FA intake could a positive value for endogenous fecal FA be obtained statistically. Depending on the equation, estimated endogenous fecal FA was 1.7 or 2.0 g/kg DMI (White et al., 2017). The alternative ap- proach of including an endogenous fecal FA term combined with true FA digestibility decreasing as FA concentration increased (White et al., 2017) was considered, but adding this complexity to the model was not necessary when the FA source classes used in the final model were included.

Assuming constant true digestibility of FA over typical FA intakes allows the digestibility of FA from an added fat supplement to be estimated from the difference in apparent FA digestibility of the basal and supplemented diet. Although the original concept of the Lucas plot was to apply to a nutri- ent that was consistent across all feedstuffs, the regression of Daley et al. (2020) extends this concept to summing FA according to fat supplement classes. The true digestion coef- ficient is constant with DMI and diet FA concentration within a class but can differ among classes. If apparent digestibility is plotted against dietary FA with no consideration of differ- ent fat classes, there are at least two reasons for this curve to have a diminishing slope. One is that the true digestibility of FA decreases with increased FA intake due to increased FA diet concentration or increased DMI. Second, if FA supple- ments with lower than average digestibility are incorporated more commonly into the higher FA diets, then the slope will also decrease with increased FA intake. The model used in the current revision can explicitly account for the latter ef- fect only. If there is an undetected general decrease in FA digestion as FA in the diet increases and if FA supplements are present in greater amount in higher FA diets, then the inherent digestibilities (see Table 4-1) of these supplements are underestimated. However, in practice, if adding the supplement increases FA in the diet and this decreases the overall FA digestibility, that effect is implicitly incorporated into the digestibility estimates in Table 4-1.

The previous NRC (2001) fat digestibility model was found to provide poor fit to the experimental data available even after fitting new parameter estimates to the existing model and allowing for an effect of DMI (White et al., 2017). White et al. (2017) derived new models to estimate fat digestibility, and all were superior to the equation used in NRC (2001) and included parameter estimates that allowed for effects of diet FA concentration and DMI on predicted FA digestibility in addition to other independent variables. However, the com- mittee chose to use the Daley et al. (2020) model because of its biological basis, its simplicity, and its overall accuracy.

Total intestinal (postruminal) apparent digestibility de- creases with increased duodenal FA flow in basal and supple-

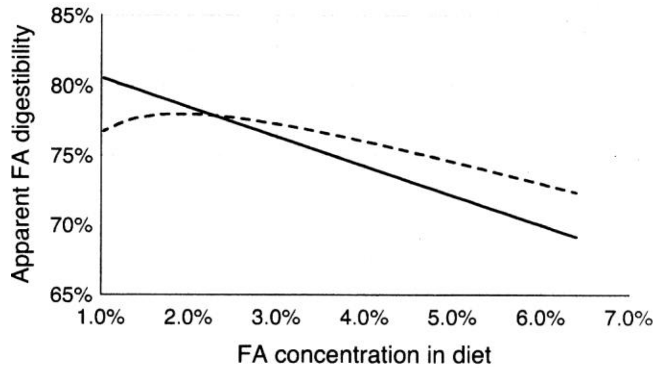


FIGURE 4-2 Predicted intestinal (postruminal) apparent digest- ibility as FA intake increases (solid line) compared to calculated total-tract apparent digestibility (dashed line).
SOURCE: From linear regression equations for duodenal flow and intestinal digestibility in Boerman et al. (2015a).

mented diets; intestinal digestibility, % = 82.5-0.0088 x du- odenal FA flow, g/d (Boerman et al., 2015a). Within that data set, duodenal FA flow ranged from 100 to 1,800 g/d, and pre- dicted intestinal digestibility decreased from 81 to 67 percent over that range. Plotting FA digestibility versus dietary FA should not be confused with a Lucas plot, although they use the same information. Under conditions of nonzero endog- enous fecal nutrient secretion and constant true digestibility, plotting apparent digestibility against intake would result in a curvilinear function rising to a horizontal asymptote as ap- parent digestibility increases with intake, approaching true digestibility at infinite intake (Palmquist, 1991). Nonzero (positive) endogenous fecal FA in conjunction with decreas- ing true digestibility could yield a curve of similar shape to the dashed line in Figure 4-2. Duodenal flow can change because of changes in DMI, FA concentration in the diet, and possible alterations in ruminal net appearance or disap- pearance of FA. As duodenal flow exceeds intake at low diet FA concentrations and is less than intake at higher FA con- centrations, the total-tract digestibility versus intake of FA will not follow the same function as intestinal digestibility. Using the linear regressions for intestinal digestibility and for duodenal flow versus intake (Boerman et al., 2015a), a calculated curvilinear function for total-tract apparent digest- ibility is obtained (see Figure 4-2). This total-tract function supports the effect of total dietary FA greater than 2 percent of diet DM continuously decreasing digestibility. However, the calculated relationship resulting from combining linear ruminal and intestinal regressions gives a relationship that is nonlinear and a smaller range of digestibility, especially for basal-type diets. This presumably explains why White et al. (2017, sup. 3) did not observe a significant quadratic response when total-tract fecal FA excretion was regressed against diet FA in low-FA diets. Previous discussion of FA intake effects on FA digestibility was based on meta-regression approaches and includes

TABLE 4-2 Effect of Amount of Supplemental Fat on Digestibility of FA in Supplemental Fat

| | Low, % of Diet DM | | High, % of Diet DM | | | |
|---|---|---|---|---|---|---|
| Fatty Acid Supplement | First Concentration of Added Fat Supplement | Fatty Acid Digestibility of Fat Supplement | Concentration of Total Fat Supplementation at Second Level of Supplement | Fatty Acid Digestibility of Fat Supplement | Fatty Acid Digestibility of Second Increment of Fat Supplement | Reference |
| Tallow[a] | 1.80 | 90.3 | 5.00 | 72.9 | 60.7 | Weisbjerg et al., 1992 |
| Mildly hydrogenated tallow | 1.80 | 39.7 | 5.7 | 52.8 | 58.8 | Drackley and Elliott. 1993 |
| Palm calcium salt | 1.60 | 93.0 | 3.40 | 87.9 | 82.8 | Weiss and Wyatt, 2004 |
| Palm calcium salt | 2.70 | 88.0 | 4.80 | 77.5 | 63.4 | Wu et al., 1991 |
| Crushed rapeseed | 2.30 | 74.7 | 4.70 | 69.7 | 65.0 | Murphy et al., 1987 |
| Hydrogenated palm | 1.7 | 38.4 | 3.4 | 36.5 | 34.5 | Weiss and Wyatt, 2004 |
| Blended fat | 2.00 | 80.6 | 3.90 | 70.1 | 75.2 | Wuet al., 1991 |

[a]This study measured intestinal digestibility (not total tract).

effects both across and within studies. However, when mul- tiple levels of the same fat supplement are added to a basal diet in the same study, digestibility of the supplemental fat decreases with increased addition (Palmquist, 1991). This was observed in five of the seven comparisons in Table 4-2. As the model of FA digestion implemented in this version ignores this effect, the user is cautioned that at very high levels of FA supplementation, overall FA digestibility and energy content of the diet may be overestimated.

The discussion and literature reviews above consider total- tract digestibility of FAs as a group. Total-tract digestibility of individual 18-carbon FAs is misleading, as biohydrogenation of unsaturated C18 FAs can lead to an overestimate of their digestibility and an underestimate for dietary C18:0 (Glasser et al., 2008b). Measuring total-tract digestibility of total Cl6 and total C18 FAs each as a unique group of FA, in addition to total FA, is possible and desirable. Changes in apparent digestibility of diet FA with added FA supplements can pro- vide an estimate of the true digestibility of the specific chain length of FA added. Drackley and Elliott (1993) observed slightly lower total-tract digestibility for C16 versus Cl8 in basal diets and with supplemental partially hydrogenated tallow. In an analysis of literature data, apparent intestinal digestibility ranged from 65.3 to 83.8 percent for long-chain FA, with a mean of 77.1 percent for C 16:0, 72.8 percent for Cl8:0, and 74.5 percent for all FAs combined (Boerman et al., 2015a). In low-fat diets, intestinal digestibility was similar at 76.7 percent for C16 and 77.8 percent for total FA. Intestinal digestibility of C18:0 was significantly lower than other C18 FAs when intestinal digestibility was deter- mined with fecal samples, although this difference appeared not to occur in diets without supplemental fat, suggesting some confounding with FA source. Intestinal FA digest- ibility declined with increasing FA intake across studies; the increased FA intake could be a result of increased diet DMI or increased FA concentration. Glasser et al. (2008b) plotted C18:0 absorption from the small intestine versus Cl8:0 duodenal flow (both per kg DMI) across ruminant species and found a decreasing absorption rate with higher duodenal flow. This negative quadratic effect did not occur for unsaturated C18 FA. Using the same data set, Schmidely et al. (2008) showed intestinal disappearance of C16 total FA as a constant 73 percent of duodenal flow, while total C18 total decreased with duodenal flow with a linear coefficient of 0.85 times duodenal flow and a quadratic coefficient of -0.0017. Based on this and other regressions in that study, when Cl8 total intake exceeds 3.8 percent of dietary DM, Cl8 intestinal digestibility fell below that of total Cl6. In- testinal digestibility of Cl8:0 drops across even low-FA diets (i.e., less than 800 g/d FA and less than 500 g/d duodenal C18:0) as FA intake increases (Boerman et al., 2015a). Ad- dition of fat sources with 33.1 percent C16:0, 53.3 percent C18:0, and 5.2 percent cis-9 C18:1, or 84.3 percent C16:0, 4.1 percent C18:0, and 8.7 percent cis-9 Cl8:1, altered the FA digestibility of the basal diet from 76.7 to 67.6 percent and 76.3 percent (Western et al., 2020). Authors calculated the digestibilities of the supplements as 55 percent and 77 percent, and most of the difference was attributed to digestion of 18-carbon FA.

Intestinal digestibility must be coupled with ruminal disappearance to predict total-tract digestibility, and in the case of the regressions from Schmidely et al. (2008), this virtually eliminates the drop in total C18 digestibility with increased dietary concentration. One additional advantage of using FA rather than CF occurs when determining digest- ibility of a FA-rich feed added to a basal diet. Generally, the non-FA CF in the basal diets is less digestible than the FA, and supplements contain most of their CF as FA. Therefore, CF digestibility may be increased with fat supplement ad- dition, even though FA digestibility may remain the same or decrease.

Digestibility of supplemental fat sources varies, so a com- mon measure that helps discern differences among fat sources would be useful. Iodine value (IV) is directly proportional to the degree of unsaturation: the higher the IV, the more unsatu- rated the fat. Low-IV feeds will usually contain considerable quantities of Cl6:0 or C18:0 or both. Unlike rumen biohydrogenation, which requires a free FA, industrial hydrogenation (partial or extensive) can be done on fatty acyl groups in intact TGs. The low IV of hydrogenated palm oil and tallow and other common fat sources is primarily the process of converting unsaturated C18 FAs to C18:0. Hydrogenated TGs from palm or animal fat, containing predominantly C16:0 and C18:0 with less than 15 percent C18:1, can have digestibility below 50 percent (Jenkins and Jenny, 1989), and increasing the degree of hydrogenation for tallow TGs (e.g., 40 to 15 percent C18:1 with corresponding IV of 45.0 to 16.4) is directly related to the decreasing digestibility (Pantoja et al., 1996). Hydrogenated tallow is less digestible than conventional tallow (Pantoja et al., 1995). Digestibility of FA in supplemental fat can be low when the IV is below 40 (Firkins and Eastridge, 1994). Data in that review suggested some benefit of increased C16 to C18, but interaction with IV and FA level made this difficult to interpret across studies. Boerman et al. (2015a) showed 08:0 intestinal digestibility to be negatively correlated to 08:0 duodenal flow, but the effect was not significant for 06:0 digestibility versus C16:0 duodenal flow.

Supplements containing mostly saturated free FAs with IV of 14 and 12 percent C18:1 were much more digestible than hydrogenated tallow triglyceride with an IV of 8 and C18:1 of 9 percent (Elliott et al., 1994), but there was no difference in FA digestibility between hydrogenated tallow triglyceride with 15.6 percent 08:1 and saturated tallow FA with 11.2 percent C18:1 (Eastridge and Firkins, 1991). In addition, supplements containing almost pure C16:0 FA can have true digestibility below 50 percent (Piantoni et al.,

2013) , and supplements that are primarily stearic can have digestibilities below 30 percent (Piantoni et al., 2015; Boerman et al., 2017). Given the sensitivity of digestibility to IV in this range, as well as uncertainty as to what specifically limits FA digestibility at elevated levels of FAs in the diet, it is not reasonable to conclude feeding saturated fats as hydro- lyzed FA or salts always removes concerns about intestinal digestibility of these fats. Ca salts of palm FA are probably the most highly digestible sources of supplemental fat and were much more digestible than strongly hydrogenated palm oil (Weiss and Wyatt, 2004). The effect is due to some combination of a relatively high IV of palm FA versus the hydrogenated palm oil (47 versus 7) due to conversion of C18:1 to C18:0 during hydrogenation; the fact that C18:1 in Ca salts of palm oil FAs are partially protected by Ca saponi- fication, thereby providing an intestinal amphiphile (Jenkins and Bridges, 2007); and the fact that the hydrogenated palm oil was a triglyceride. Predicting the digestibility of a fat source from its FA content and esterification state is risky, and having empirical estimates of digestibility, especially for extensively saturated FA sources, seems prudent.

Decreasing particle size of dry granular fats may increase digestibility, but responses have tended to be small and not

statistically significant. A summary of trials (Firkins and Eastridge, 1994) indicated that mean FA digestibility of prilled (n = 8) and flaked (n = 5) hydrogenated tallow was 77 and 69 percent, respectively. Fat structure, the form in which FAs are fed, may have modest effects on digestibility. A re- view of the literature (Firkins and Eastridge, 1994) indicated that FA digestibility of diets containing triglyceride prills or FA prills was 77 or 73 percent of control diets without added fat. However, effects of fat structure might have been confounded: mean IV and Cl6:18 ratios were 20.7 and 0.41 for triglyceride prills and 11.2 and 0.45 for FA prills. Mean prill sizes between 284 and 325 microns had no effect on FA digestibility, but 600-micron prill size increased total-tract digestibility of total, C16, and C18 FAs compared to smaller prills when an 85 percent palmitic acid fat supplement was fed (de Souza et al., 2017).

Direct estimates of total C16 and C18 FA digestibility in basal and supplemented diets to derive an empirical estimate of the digestibility of an FA supplement should be obtained when manufactured supplements containing mostly fat are used. Ideally, the fat supplements should be tested at two levels above basal to quantify any digestion depression. The same is true of full fat seeds in various mechanically pro- cessed or whole forms.

Effects of fat sources on DMI must be considered when assessing the value of supplementation on energy intake. In addition, if these dietary fats contain unsaturated FAs that induce milk fat depression and reduce milk fat secretion, retained tissue energy balance may be positive even if energy intake is not increased (Harvatine and Allen, 2006). Adding a fat with less than half the digestibility of the carbohydrate it replaces will not increase the DE density of the diet much, but it does decrease the fermentation load in the rumen, which may be an advantage. Replacement of starch with fiber and a palmitic acid-based fat supplement resulted in equal energy intake with enhanced milk energy output; this effect could be useful to reduce body condition gain in later lactation cows (Boerman et al., 2015b).

Oilseeds are rich in unsaturated FAs (Glasser et al., 2008a); therefore, FA profile as fed is not a major source of varia- tion for FA digestibility. Digestibility of FA in oilseeds is probably more a function of the size and physical nature of the specific oilseed and subsequent processing. If oilseeds contain FAs, especially linoleic acid, that are precursors for bioactive FAs, fine processing should be avoided to reduce milk fat-depressing effects, and this lack of processing might then reduce FA digestibility. In lactating cows, intact oilseeds have lower intestinal digestibility of FA than other FA sources, but this can be improved by grinding (Boer- man et al., 2015a). Delinting of cottonseed reduces its FA digestibility, but mechanical processing can enhance biohy- drogenation intermediates that depress milk fat (Reveneau et al., 2005). In steers, FAs in whole canola seed are poorly digested, but this is improved by processing (Aldrich et al., 1997). However, even crushed rapeseed supplementation reduced FA digestibility compared to the basal diet (Murphy et al., 1987). Reducing the particle size of roasted soybeans did not affect total-tract FA digestibility (Tice et al., 1993).

Final assignment of digestibility coefficients to FA sup- plied by different feeds was based on regression analysis of 30 published studies that provided total-tract apparent digestibility of FA and diet FA information (Daley et al., 2020). Study was considered a random effect, and observa- tions were weighted using reported standard errors. Intakes of apparently digested FA were regressed on FA intakes provided from 11 classes of feeds using feed library infor- mation and reported dietary FA information to yield true digestibility coefficients (see Table 4-1). Several studies included fat supplements that were poorly digested, and these were either TGs highly enriched in saturated FA or almost pure palmitic or stearic FA supplements. Not all fat supplements included in the regressions have example feeds included in the printed or electronic feed tables included in this report; however, their presence in the regression model is required to better estimate FA digestibility coefficients for other FA sources included in the library. The FA composi- tion was used to provide nonoverlapping classes of diverse fat sources while having at least some replication within a fat class across multiple papers. These classifications do not necessarily extend beyond that database. The one example of very highly enriched palmitic acid was grouped with two examples of highly enriched stearic acid primarily to exclude them from other classes as all three had digestibilities that would probably preclude their use as commercial supple- ments. The slightly greater digestibility of oils over tallow can be expected based on FA profile, but for reasons that are not clear, the value for blended animal/vegetable fats did not fall in this continuum. This total-tract FA digestibility data set includes papers reporting intestinal digestibility plus those reporting only total-tract apparent digestibility. In this larger data set, neither diet FA concentration nor DMI significantly improved model fit of total-tract apparent FA digestibility, and these terms were not included in the final model used to derive FA digestibility of the FA classes. This regression also included a fixed zero intercept. Not fixing the intercept to zero resulted in a nonsignificant positive digestible FA intake at zero FA intake, corresponding to an impossible negative endogenous secretion. This set of conditions provided optimal fit and was consistent with zero endogenous FA secretion, true digestibility, and apparent digestibility being equal and constant true digestibility as FA intake increased.

## MILK FAT COMPOSITION

Milk fat has a much more complex FA composition than dietary fat. Milk contains more than 400 different FAs, but only about 14 are present above 1 g/100 g milk fat (Jensen, 2002). Milk FAs can be derived from de novo synthesis in the mammary gland, from FAs synthesized in the adipose tissue and subsequently mobilized or absorbed FAs of dietary or microbial origin (Glasser et al., 2008a; Shingfield et al., 2010). Milk FAs synthesized in the mam- mary gland are primarily C4:0, C6:0, C8:0, C10:0, C12:0, C14:0, and C16:0. The six FAs shorter than C16 form about one-quarter of milk FAs by weight, but a larger fraction of milk FA if expressed on a molar basis. Mammary gland synthesis of C16:0 is augmented by dietary supply as many feeds and fat supplements contain Cl6:0, so milk C 16:0 is a combination of synthesized and absorbed FAs. Milk Cl6:0 yield, which is usually about one-third of milk FA yield by weight, can approach or even exceed 50 percent of milk fat by weight when supplementary Cl6:0 is supplied (Loften et al., 2014). Milk yield of Cl6:0 is therefore dependent on the overall rate of mammary gland C16:0 synthesis, blood supply of dietary Cl6:0 and Cl6:0 supplied by the adipose tissue (minus any desaturation into Cl6:1), and the efficiency of their incorporation into milk triglyceride. Most of the rest of the fat in milk are 18-carbon FAs derived directly from the diet but can come from body tissue. Most C18 in milk is oleic acid (20 to 30 percent of total milk FA by weight) followed by Cl8:0 (9 to 14 percent; Jensen, 2002). As explained previously, most of the absorbed C18 will be Cl8:0 and Cl 8:1. Milk 18:1 can betaken up directly from blood or produced from blood-derived C18:0 by ac- tivity of delta-9-desaturase in the mammary gland. Oleic acid content of milk is increased by body fat mobilization (Jorjong et al., 2014). For reasons that are not entirely clear, very little Cl6:0 or shorter FAs are desaturated, so milk has only small amounts of monoenoic FAs shorter than C18/:1 (about 3 percent of total milk FA by weight). Milk TGs contain a mixture of FAs within a molecule. Based on carbon number (total number of FA carbons per triglyceride molecule), milk TGs range from 28 (e.g., this could be a triglyceride with C4, C10, and C14 or one with C4, C8, and C16 acyl groups esterified to the glycerol) to about 54 (which would be a TG with three 18-carbon fatty acyl groups). Physical characteristics of milk TGs may be impacted by total FA proportions and by the distribution of FAs with the various TGs. TGs with different FA residues on the noncentral carbons of glycerol have an anomeric center at the glycerol number 2 carbon, and therefore the 1 and 3 positions are distinguished as snl and sn3. Many milk TGs contain at least one short-chain FA, preferentially bound to the sn-3 position (Parodi, 1982). Although milk FA composition is generally reported on a weight basis, TGs composed of one butyryl, one palmityl, and one oleiyl moiety (carbon number 38) are equimolar in these FAs (33.3 percent each); however, on hydrolysis, this example TG would yield 14 percent butyric, 41 percent palmitic, and 45 percent oleic by FA weight. For the major FAs present in milk, coefficients of variation may be up to 15 percent, while minor FAs have a greater variation relative to their absolute amount.

## LACTATING COW RESPONSES TO DIETARY FAT

Production responses to supplemental fat are dependent on the nature of the fat and lactation stage. Milk yield response to supplemental fat is influenced by several dietary factors, including basal diet, fat composition of the supplement, and amount of supplemental fat, as well as animal factors such as stage of lactation, energy balance, and level of produc- tion. Milk fat yield response is best divided into secretion of milk FA that can be synthesized in the mammary gland (C16 and shorter, primarily saturated) and those that reflect incorporation of mobilized adipose or dietary FA into milk (C16 and C18, with the largest proportion of the C18 secreted as oleic). Comparing yields of these milk FAs to determine shifts in milk FA precursors is preferable to comparing the proportion of each in total FA, as any increase in the propor- tion of one group necessitates a decrease in the proportion of another, even if yields of the FA with lowered proportion are not decreased. Adding dietary C16:0 increases milk fat yield due to greater secretion of milk C16:0, although the apparent marginal efficiency of transfer is approximately only 15 to 35 percent (Lock et al., 2013; de Souza et al., 2016; Dorea and Armentano, 2017). If C16:0 is fed versus 08:0, yield of C16 and C18 FA in milk responds in kind (Rico et al., 2014) . Actual transfer of dietary C16:0 to milk 06:0 may be greater than apparent efficient transfer if the added dietary 06:0 depresses mammary de novo 06:0 synthesis.

Formation of FAs that are bioactive and milk fat de- pressing is enhanced by rapid carbohydrate fermentation and corresponding low ruminal pH and requires a source of unsaturated FA precursors (Griinari et al., 1998). Both these conditions can occur in lactating cows consuming large amounts of typical mixed rations with low dietary FA concentration (Stoffel et al., 2015). At these high levels of DE intake, even the low, native unsaturated FA content of common dairy forages and concentrates may be adequate to produce enough bioactive FAs to cause inhibition of de novo synthesis of milk FA. In addition, dietary cation-anion balance of the diet also modifies the potential for milk fat depression (see Chapter 7).

Dietary Cl6:0 presumably produces no bioactive FA to reduce secretion of de novo milk FA and has little effect on total mass secretion of FA with 14 carbons or less or secre- tion of total milk 18-carbon FA (Lock et al., 2013; Dorea and Armentano, 2017). Added dietary C16:0 does, however, shift de novo milk FA (including de novo C16) to shorter chain lengths and lower molecular weight (Enjalbert et al., 1998), so unchanged mass secretion (g/d) implies increased secre- tion on a mol/d basis. Feeding palmitic in place of stearic acid also caused a linear reduction in secretion of milk FAs as carbon chain length increased from C8 to C14 (Rico et al., 2014). Similarly, dietary C18:0 should increase milk C18:0 and C18:1 without reducing secretion of shorter milk FAs through generation of bioactive FAs. Eighteen-carbon un- saturated FAs can decrease secretion of de novo FAs through generation of bioactive FAs but may elevate secretion of milk C18 FAs, mostly as oleic and rrans-monoenoic FAs (Stoffel et al., 2015). The net result on total milk fat secretion when adding unsaturated C18 FAs is dependent on the balance of these opposing changes. Typically, milk with an FA profile favoring C18:1 and C18:0 is increased at the expense of shorter saturated FAs synthesized exclusively in the mam- mary gland (Glasser et al., 2008a). High levels of added linoleic acid decrease total yield of milk 18-carbon FAs (He et al., 2012), presumably due to synthesis of active milk fat-depressing FAs such as trans-10, cis-12 CLA (Jenkins and Harvatine, 2014). Oleic acid can reduce milk fat synthesis but is less potent than linoleic (He et al., 2012). This negative effect of oleic may be due to synthesis of bioactive FAs in the rumen, but oleic is not known to be converted to any CLA. The active FA may be trans-10 18:1 or some other bioactive monoenoic FA, or oleic acid could indirectly increase CLA formation from PUFA. When compared directly, linolenic acid has not proven more deleterious than oleic (Kelly et al., 1998; AbuGhazaleh et al., 2003; Rego et al., 2009), and in some studies, it is less milk fat depressing than linoleic (He and Armentano, 2011). A review of studies on feeding oils and oilseeds concluded that both linoleic and linolenic FAs reduced shorter-chain milk FAs compared to milk 18-carbon FA (Glasser et al., 2008a). A regression analysis across di- etary FA concentrations confirms that unsaturated dietary FAs cause a linear depression of de novo milk FAs (Dorea and Armentano, 2017). Palm oil (not to be confused with palm kernel oil) and tallow are both mixtures of Cl6:0 and C18:1, although tallow has more C18:0 and less C16:0 than palm oil. These fats behave as expected, with increased C16:0 secre- tion from absorbed C16:0, decreased secretion of C14:0, and shorter FAs due to the mild inhibitory effects of dietary oleic acid and small amounts of PUFA, as well as small increases in secreted milk C18 due to absorbed C18 FAs. This would explain the generally positive effects of tallow and palm oil on milk fat yield whether fed as TGs, FAs, or Ca salts.

Infusion of FAs or TGs into the abomasum or duodenum allows the effect of FAs to be measured independently of ruminal microbial alteration and ruminal synthesis of bioac- tive FAs. In these experiments, the range of infusion has been approximately 200 to 600 g/d. If the typical flow of linoleic acid at the duodenum is about 100 g/d and around 15 g/d for linolenic acid, then for some of the more unsaturated fat sources used, the levels of C18:2 or C18:3 infused may be quite high compared to normal. Results have been mixed. No decrease for milk short-chain FAs was seen when fat low in PUFA was infused (Enjalbert et al., 2000; Kadegowda et al., 2008). Studies using TGs or FA blends richer in PUFA have produced variable results. Infusion of linseed oil re- sulted in no change in C16 and shorter milk FAs secretion but increased longer-chain milk FA yield (Moallem et al., 2012). Christensen et al. (1974), Drackley et al. (1992), and Litherland et al. (2005) all reported some decrease in various milk short-chain FAs with fat infusion. The three latter studies also decreased DMI with infusion of fat, which may be an important causative factor in the decreased milk FA synthesis. Nevertheless, the potential for some postruminal inhibitory effect of the unsaturated FAs on milk fat secretion cannot be totally discounted.

If fat supplementation is started during the early post- partum period, a lag may occur before a positive milk response is observed (Jerred et al., 1990; Schingoethe and Casper, 1991). An extensive summary by Chilliard (1993) indicated that the average fat-corrected milk response to fat supplementation (average increase of 4.5 percentage units in dietary CF) during early lactation (beginning before 4 weeks postpartum and ending before 11 weeks postpartum) was 0.31 kg/d and not significantly different from controls. Average 4 percent fat-corrected milk response to fat supple- mentation during peak lactation (beginning before 8 weeks and ending by 24 weeks postpartum; average increase of 3.6 percentage units of dietary CF) was 0.72 kg/d and sig- nificantly different from controls. In middle to late lactation (average increase of 3.4 percentage units of dietary CF), the response was 0.65 kg/d, which was not significantly differ- ent from control. Average milk production of cows in this summary was less than 35 kg/d. Milk-yield responses to supplemental fat in cows that produce more than 40 kg/d are not well defined. Using cows with prior production ranging from 32.2 to 64.4 kg/d, response of milk production and DMI to added stearic acid fat source was greater for the more productive cows (Piantoni et al., 2015). Some of the variation in response of milk protein and fat yields to added fat may be due to varying depression of feed intake when feeding different supplemental fats.

Nutritionists often apply a maximum to the amount of unsaturated FAs in lactating cow diets to prevent the onset of milk fat depression. There are three reasons to question this concept. First, bioactive FAs have their greatest incremental effect at the lowest level, and the effect diminishes as the concentration of these FAs increases further in milk (Bau- man and Lock, 2006). Second, adding fat to the diet clearly depresses de novo FA formation while raising secretion of preformed dietary FAs in milk. The effect of depression of FA synthesis in the mammary gland by dietary unsaturated FA may be masked (but still present) at low levels of FA ad- dition when only total milk fat concentration is measured. This could be detected by milk FA analysis while it would be missed by measuring only total milk fat concentration. Fi- nally, the effect of unsaturated FAs on reducing milk fat syn- thesis in the mammary gland happens at total diet FA below 3 percent, which can be achieved by adding 1.7 percent oil to basal diets containing only 1.2 percent FA (Stoffel et al., 2015) with a linear affect across dietary FA concentrations (Dorea and Armentano, 2017). The effect at these very low levels of FA may or may not apply to FA in basal feedstuffs, but that is difficult to determine experimentally without in- troducing confounding factors. Grasses differ in FA content and composition based on species and maturity, with total unsaturated FA content more than 2.5 percent possible (Mir et al., 2006). Pasture with more than 4 percent FA is not uncommon (Roche et al., 2011). Based on a summary of FA concentrations (Kalac and Samkova, 2010), perennial ryegrass can range from 2.2 to 4.4 percent total FA with 1.1 to 3.2 percent linolenic acid, and corn silage can have 1.2 to 4.0 percent FA. Following linolenic acid, linoleic and palmitic acids are the next most prominent FAs in leafy forages, but forages containing grains and seed may also contain oleic acid. Even in diets not containing exogenous fat supplements, variations in feed FA concentration and type of FA may alter milk fat secretion. Increased concentration of unsaturated diet FAs may lower the ratio of milk FAs with carbon chains of 16 or less to milk FAs with C18, and shifts from PUFA to saturated FAs in feeds may increase total milk fat yield, although both of these effects are likely to be small.

Milk fat and total milk yields often increase in response to fat feeding, but protein concentration decreases. Milk protein concentration can decrease due to decreased secre- tion of protein or because of dilution by increased lactose secretion and volumetric milk yield. The effect of dietary fat to reduce milk protein concentration diminishes slightly as the amount of supplemental fat increases: for example, $y = 101.1 - 0.638x + 0.0141x^2$, where y = milk protein con- centration [(treated/control, percent) x 100] and x = percentage of total dietary fat (Wu and Huber, 1994). Casein is the milk nitrogen fraction that is most depressed (DePeters and Cant, 1992). Although milk protein percentage is usually depressed, protein yield usually remains constant or is increased. Of 83 comparisons (fat supplementation versus control) summarized by Wu and Huber (1994), milk protein yield was unchanged or increased in 57 comparisons and decreased in 26. However, in 15 of the 26 comparisons in which protein production was de- creased, milk production also was decreased. A meta-analysis restricted to continuous lactation trials showed decreased protein percentage, increased milk yield, and no change in milk protein production even though DMI declined with fat addition to the diet (Rabiee et al., 2012). Adding fat supple- ments containing mostly C16:0 and Cl8:0, Hu et al. (2017) reported increased yield of milk, milk fat, and milk protein, while protein percentage usually declined. Why milk protein production does not increase at a similar rate to milk volume with fat supplementation has not been determined.

## DIETARY FAT INTERVENTIONS AND REPRODUCTION

Fat supplementation can positively influence reproductive performance of dairy cows. A summary of 20 studies indi- cated that first-service conception rate or overall conception rate was increased in 11 of the studies (Staples et al., 1998). Over all studies, the mean increase was 17 percentage units. Three studies found a negative influence of supplemental fat on reproduction, but the effects were confounded by substantial increases in milk production. Feeding fat increases fol- licle numbers and the size of the dominant follicle. Whether those changes in follicular dynamics have a positive effect on reproductive performance is unknown. Potential mechanisms by which fat influences reproduction include amelioration of negative energy balance, enhancement of follicular develop- ment via changes in insulin status, stimulation of progester- one synthesis, and modification of the production and release of prostaglandin F2a (Staples et al., 1998). In the studies reviewed by Staples et al. (1998), change in energy status was not related to change in conception rate. Likewise, the effects of fat on circulating insulin have not been consistent, although the trend is toward a reduction. How a reduction in plasma insulin could benefit reproduction has not been determined, but the impact of glucose balance during lacta- tion and its interaction with reproductive processes has been reviewed (Lucy et ah, 2014). Fat supplementation consis- tently increases plasma progesterone concentration, but the change might be because of depressed clearance rather than increased production (Hawkins et ah, 1995). Staples et ah (1998) proposed that feeding fats that are rich in linoleic acid suppresses prostaglandin F2a and prevents regression of the corpus luteum. The importance of considering omega-3 and omega-6 FA was reviewed by Santos et ah (2008). Thatcher et ah (2011) proposed mechanisms whereby the proper ratio of duodenal linoleic and linolenic acids could benefit reproductive performance. Effects of supplementing omega-3-rich FA sources in the diet to favorably influence various reproductive processes and overall reproductive success are reviewed in Moallem (2018). A more recent meta-analysis also concluded that changes in fat nutrition improved gross reproductive performance (Rodney et ah, 2015). However, the manipulations summarized as treatments in this analysis included changes in total dietary FA from various sources, addition of CLA, and feeding isolipid diets where the pattern of FA changed, so the nature of the treatment is not clear except that it involved changes in some aspect of fat feeding. For example, one treatment was substitution of flaxseed for sunflower seed (increased linolenic acid), and another treat- ment was substitution of Ca salts of palm oil for flaxseed (decreasing linolenic). One of the most common treatments was mixed CLA (de Veth et ah, 2009), which improved re- production when supplemented up to 10 g/d. Interestingly, this effect was not related to improved energy balance caused by reduced milk FA yield. Given the different patterns of bioactive FAs formed when different unsaturated C18 FAs are fed, this effect of CLA supplements supports the need for measuring FA effects independently as opposed to total dietary FA content.

The difficulty of summarizing these data into a clear and actionable effect is obvious. An extensive review is available (Roche et ah, 2011) of the effects on reproductive function in lactating cows of dietary interventions, including modifying FA amounts and FA profile. These authors stated that the effect of dietary fat on reproductive outcomes “is difficult to interpret,” and the most impactful nutritional management should be directed toward achieving optimal body condition in freshening cows. Berry et ah (2016) indicated that while evidence for nutritional effects on reproduction existed, the scale of the effect was probably commonly overstated. In addition, many studies of nutrition and reproduction fail to properly identify the experimental unit and may incorrectly interpret replication. Large numbers of animals are required for accurate estimates of gross reproductive performance, and recommendations for effective designs have been made (Lean et ah, 2016).

## DAIRY FAT AND HUMAN HEALTH AND POSSIBLE MODIFICATION OF MILK FAT

Dairy products are processed and can have fat removed or concentrated relative to raw milk. In addition, feeding methods and genetics of dairy animals can alter raw milk fat concentration and FA composition. Most often, dairy fat is eaten as part of dairy foods that contain nonfat components that may also impact human health, positively or negatively, which readers need to recognize. The most recent U.S. government guidelines for human diets identified Ca and vitamin D as shortfall nutrients and saturated fat as being overconsumed (HHS and USDA, 2015). Studies that directly measure human health and longevity are epidemiological in nature and, therefore, can never conclude cause and effect, only association. When human studies utilize classical dietary intervention, the end point is almost always measurement of a biomarker that, in turn, has reputed association with sub- sequent long-term health results. Studies may specifically study the effect of milk fat on human atherogenic disease (or a biomarker) or be more general and study consumption of the “typical” range of dairy products and overall human health. A recent study of this latter sort suggests that dairy food consumption is not associated with all-cause mortality and has positive associations with several important health mea- sures (Thorning et al., 2016). Considerable disparity exists in epidemiological reviews and summaries regarding the impact of dairy products on human health, and it is beyond the scope of this chapter to review them. However, even if dairy foods are in fact beneficial, this does not preclude the possibility that milk products, and specifically milk fat, could be altered to make milk products more beneficial for human health.

Human health could potentially be improved by modi- fying milk fat via feeding by (1) increasing the content of various beneficial bioactive FAs present in milk in trace, but potentially effective, concentrations; (2) altering the major FA composition of milk to reduce saturated FA and elevate oleic or PUFA while maintaining or increasing milk fat con- centration; (3) reducing total milk fat yield relative to fluid and protein, which in tum could be used to produce lower-fat dairy products (or human diets); and (4) enriching the content of milk omega-6 or omega-3 FAs.

Milk contains over 400 FAs; many of these are in trace amounts. Some of these FAs can have potent biological ef- fects (Lock and Bauman, 2004). This bioactivity could plau- sibly result in various positive and negative effects on human health, many of which may not be related to plasma choles- terol. The branched-chain FAs in milk mostly derived from microbial sources were deemed underexplored for human health (Taormina et al., 2020). Epidemiological studies of milk products have indicated positive health associations with dairy products, but the role of the specific dairy foods and dairy fat component is not clear (Elwood et al., 2010). The actions of different CLAs on human health are complex. Changes in the diet of cows can cause diverse changes in the trace FAs present in milk, and often many of the trace FAs are not measured accurately. These complexities prevent any clear road to enhancing potential health benefits by us- ing nutrition to alter the trace bioactive FAs present in milk. However, the potential positive effects of these FAs should not be discarded when considering the recommendations for the level of milk fat in the human diet.

The effects of dairy products and dairy fat on human health are not resolved, but it is clear that the general advice for extremely low-fat, high-carbohydrate diets (no more than 20 percent of calories from fat) was not well justified at its in- ception and is being abandoned by experts in human nutrition (German and Dillard, 2004). Broad, but not unanimous, con- cerns remain over too large a proportion of human dietary fat coming from saturated fat (Burlingame et al., 2009; USDA, 2015). The current trend in human dietary recommendations is maintaining fat levels at about 30 percent of calories but increasing MUFAs and PUFAs, while reducing medium-chain saturated FAs and industrially hydrogenated fats due to their content of various trans-FAs. In addition, common guidelines still often call for consumption of “low-fat” dairy products because of beneficial effects of dairy products but concerns that dairy fats do not help achieve the desired FA profile for the human diet. Most evidence suggests health benefits of milk for children, with limited evidence to sup- port decreasing milk fat (O’Sullivan et al., 2020) or milk fat globule membrane components (Ortega-Anaya and Jimenez-Flores, 2019). Consuming milk or milk products, regardless of their saturated FA content, was associated with positive outcomes on cardiovascular health (Mena-Sanchez et al., 2019; Rietsema et al., 2019; Companys et al., 2020; Hirahatake et al., 2020) and metabolic syndrome (Drehmer et al., 2016; Mena-Sanchez et al., 2019).

The broad terms “saturated fat” and “animal fat” persist in relation to human consumption and labeling of foods but are of limited utility in describing milk fat, which contains saturated FAs ranging from 4 to 22 carbons in length. Satu- rated FAs of different chain lengths have markedly differ- ent effects on typical biomarkers used to assess potential atherogenic risk. Consumption by humans of lauric, myristic, and palmitic acids (C12:0, C14:0, and C16:0) is related to elevated low-density lipoprotein (LDL) cholesterol, a bio- marker for negative coronary health impacts (Kris-Etherton and Yu, 1997). Shorter (Cl0:0 and lighter) saturated FAs, as well as Cl8:0, stearic acid, appear neutral relative to con- suming carbohydrate, while C18:1 has positive (cholesterol- lowering) effects (Kris-Etherton and Yu, 1997). Controlled substitution of these medium-chain saturated FAs for carbo- hydrate clearly elevates LDL cholesterol, but C12:0 simulta- neously reduces the total to high-density lipoprotein (HDL) cholesterol ratio, which is a marker associated with better cardiovascular health (Mensink et al., 2003). Therefore, the primary putative negative human health claim associated with dairy fat is most likely limited to the effect of medium- chain saturated FAs on human LDL cholesterol, which seems to ignore the neutral or beneficial effects of these same FAs on total to HDL cholesterol and ignores differences among lauric, myristic, and palmitic acids. Whether these changes in LDL cholesterol from these FAs translate into higher risk for human cardiovascular disease and decreased longevity when consuming dairy products containing fat is far from established. Generalizations are complicated by the form of dairy fat consumed and the possible positive impacts of dairy products on HDL, blood pressure, and type 2 diabetes (German et al., 2009).

The FAs fed to dairy cattle affect the FA proportions in dairy food fats. Feeding Cl6:0 fats to cows increases total milk fat secretion mostly by increasing Cl6:0 with only slight absolute decreases in Cl4:0 and Cl2:0, with predict- able increases in the molar proportion of Cl6:0 in milk relative to other FAs. As described earlier, increased C18 unsaturated FAs in lactating cow diets will reduce C16 and shorter chain saturated FAs in milk while usually elevating C18:0 and C18:1 in milk fat. Therefore, a simple strategy to alter milk FA composition to the suggested healthier bal- ance of more Cl 8:0 (neutral) and Cl8:1 (beneficial) and less medium-chain (C12:0 through 06:0) FAs is to include oils that contain oleic, linoleic, and linolenic acids in lactating cow diets, which reduces yield of milk short-chain FAs while increasing milk 18-carbon FA secretion. Added stearic acid in the dairy diet may not always decrease secretion of short- chain milk FAs but can increase secretion of milk 18-carbon FA, so it too can alter the milk FA proportion in this same direction. Feeding more unsaturated FAs to cows, however, can also increase trans-FAs in milk. The effect of feeding cows to alter dairy fats in this manner and then subsequently feeding this modified fat to humans has been reviewed by Livingstone et al. (2012). Careful evaluations of the data present in the original studies reveal that three of five stud- ies found statistically significant reductions of total blood cholesterol; three showed significant reduction in LDL cholesterol, and only one study showed a significant change in HDL cholesterol, which was increased by the modified milk fat (Noakes et al., 1996; Tholstrup et al., 1998, 2006; Poppitt et al., 2002; Seidel et al., 2005). Overall, these data suggest a tendency for LDL and the total cholesterol to HDL cholesterol ratio to be lowered by these fats, a potentially beneficial change in milk fat relative to human atherogenic health, although the data are neither extensive nor completely consistent.

There are several caveats to feeding vegetable oils to lactating animals to modify dairy fat. Excess oil feeding to cows can reduce total milk fat secretion and concentration, reducing the value of milk under current marketing schemes. Although milk-testing labs can estimate the FA profiles of milk fat at the farm level to use in payment schemes and incentives, there is currently no widespread economic incen- tive in the United States to alter milk FA profile, but there is for elevated milk fat concentration and yield. Feeding veg- etable oils not only increases milk oleic acid while reducing medium-chain FAs in milk but also may increase flow of 18-carbon trans-monoenoic FAs into the duodenum followed by absorption and secretion in milk. Even if trans-FAs from ruminants may be less of a health concern than the trans-FAs from industrially hydrogenated TGs (Gebauer et al., 2011; Oteng and Kersten, 2019), current food labeling does not distinguish these and simply reports total trans-FAs. This type of oil feeding can elevate trans-FAs above 5 percent of milk fat, which could cause trans-FAs to exceed 0.5 g in a 10-g serving of dairy fat. The Food and Drug Administration in the United States requires labeling trans-FA content when the 0.5-g per serving level is exceeded. Recommendations from FAO/WHO called for less than 1 percent of human en- ergy consumption from trans fats from all sources, ruminant and industrial, combined (Burlingame et al., 2009). Dietary unsaturated FAs, especially linoleic acid, even at low levels present in basal diets, will often shift the milk FA profile to a higher content of Cl8:1 (including trans-C 18:1) and lower medium-chain saturated FAs without noticeably changing milk fat yield. Because measurement of the milk FA shift is practical with existing milk analysis procedures, an economic incentive to alter milk FA profile to achieve this shift is pos- sible. However, the benefit of such a shift in milk FA profile to human health is not proven, and labeling requirements for trans-FAs must be considered.

Factating cow diets that promote lower milk fat through highly fermentable carbohydrate and less effective fiber can yield low-fat milk products that can be left low fat or fortified with nondairy fat with different FA profiles. These diets tend to reduce milk FAs of all lengths. Use of bioactive milk fat synthesis inhibitors, such as trans-10, cis-12 CLA, can have the same effect and potentially avoid the negative animal health effects and reduction in fiber digestion associated with subacute rumen acidosis due to rapidly fermentable carbohydrate and shorter fiber. Direct feeding of these potent bioactive CLAs to cows inhibits milk FA secretion and low- ers short-chain saturated FA relative to C18:1 and total C18 (Bauman et al., 2011) without elevating trans-FA content of milk that occurs when increased amounts of unsaturated FAs are fed as a source for synthesis of the milk fat-depressing CFA (Perfield et al., 2007). However, the demand for milk fat and the economic incentives for milk fat production through milk fat pricing have not moved dairy production in this direction.

Slight increases in PUFA in milk are possible by altered feeding of cows with current technologies but will not greatly increase human dietary intake of these PUFAs. Content of linoleic and linolenic acid in milk is minor and difficult to change in unprocessed milk by feeding methods for reasons discussed above and reviewed by Lanier and Corl (2015) and Lock and Bauman (2004). For example, milk fat on average has about 0.6 percent of FA as a-linolenic acid. If a human diet derived 30 percent of calories from fat and all that fat was dairy fat, then 0.18 percent of calories would be from lino- lenic acid. Doubling this linolenic milk content, which has been accomplished by linolenic supplementation of cow di- ets, would deliver about 0.36 percent of human calories from linolenic acid. This would still be below the lower end of the recommended range of 0.5 to 2 percent of calories from linolenic acid. In contrast, ubiquitous soybean oil contains 8 percent linolenic acid, so the same increase (0.18 percent to 0.36 percent human calories from linolenic acid) could be obtained by human consumption of 27.5 percent calories from dairy fat and 2.5 percent calories from soy oil. In addi- tion, many rich sources of linolenic acid also contain higher ratios of desirable eicosapentaenoic acid and docosahexae- noic acid to linolenic. Dairy processing can easily remove fat from dairy products to yield low-fat products. When high-fat “dairy” products are desired, dairy fats can be combined with other oils or specific FA supplements to deliver the desired FA in significant amounts, while avoiding the biological inef- ficiencies inherent in the lactating cow. These changes affect melting point and other important properties of the fats and can be much more precisely standardized in food processing than by animal feeding.

## SUMMARY

For maximum performance, cows generally should not be fed diets with more than 7 percent of the DM as FA, and the economic optima may be considerably less. Feeding higher concentrations of fat can result in reduced DMI, even if the fat has minimal effects on ruminal fermentation and fiber digestion (Schauff and Clark, 1989; Weld and Armentano, 2017). A reduction in DMI may negate part or all of the advantage of using fat to increase dietary energy density and can limit milk-production responses. Optimal amounts of fat to include in diets depend on numerous factors, including the FA composition of the fat, physical and chemical form of fat, the basal diet, stage of lactation, environment, level of milk production, and feeding management. Limiting dietary FA concentration to less than 5 percent might be prudent during early lactation, when feed-intake depression due to fat supplementation has been observed (Jerred et al., 1990; Chilliard, 1993). Cereal grains and forages can contain about

3 percent FA, but FA profiles can vary significantly. Oilseeds and animal or animal-vegetable blends are acceptable fat supplements; however, at high levels of supplementation, choice of FA supplements needs to be carefully balanced to prevent milk fat depression or reduced DMI.

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